

A Web-based Flea-Market System for Strathmore University Students

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Declaration

We declare that this work has not been previously submitted and approved for the award of a degree by this or any other University. To the best of our knowledge and belief, the research proposal contains no material previously published or written by another person except where due reference is made in the research proposal itself.

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Abstract

Informal peer-to-peer commerce among students plays a vital role in campus life, enabling the exchange of academic materials, electronics and daily essentials. However, these transactional activities rely on fragmented, informal communication channels such as social media groups or physical notice boards. This lack of a dedicated framework creates significant market inefficiencies, a lack of price transparency and difficulties in coordinating physical meetups due to strict academic timetables. To address these challenges, this study proposes the development of a localized, web-based flea market system tailored specifically to the university ecosystem.

The primary objective of the proposed system is to centralize and streamline campus commerce by introducing a structured digital ecosystem. The platform will be engineered using a stable three-tier architecture comprised of a presentation layer, an application processing layer and a relational data storage layer. On the client-side, the user interface will be developed utilizing standard web technologies, specifically HTML5, CSS3 and JavaScript, to ensure a fully responsive, mobile-first consumer experience. The backend application logic and secure session management will be driven by server-side PHP scripting while user profiles, structures classified advertisements and scheduling details will be managed through a normalised MySQL relational database. To ensure 24/7 system availability and automated continuous deployment, the entire platform build will be hosted on Render's cloud infrastructure. Ultimately, the system aims to transition the campus flea market from an uncoordinated informal activity into an efficient digital marketplace, effectively saving time for students and empowering student entrepreneurs withing the university.

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List of Abbreviations

RDBMS – Relational Database Management System

OOAD – Object-Oriented Analysis and Design

Chapter 1: Introduction

1.1 Background

Institutions of higher learning provide environments that extend beyond academic instructions by facilitating social and economic interactions among students. Within these environments, informal commercial activities frequently emerge as alternative mechanisms through which students access affordable goods and services. Podkalicka and Fredriksson (2023) explain that informal and community-based marketplaces continue to play a significant role in supporting localised economic participation and exchange, particularly among young populations and learning communities. Digital and hybrid marketplace ecosystems have increasingly transformed the way individuals exchange material goods by integrating physical market activities with online communication platforms.

Globally, flea markets and second-hand marketplaces have evolved from traditional physical trading spaces in two digital mediated marketplaces that combine physical interaction with online information sharing. Modern marketplace systems are now heavily influenced by digital technologies that support communication, visibility of goods, and accessibility of sellers to potential buyers as observed by Podkalicka and Fredriksson (2023). These developments have enabled marketplaces to become more organised and accessible while preserving their role as affordable alternatives to formal retail systems.

Recent research on online and second-hand marketplaces demonstrates a growing reliance on digital platforms for the exchange of goods. A 2023 exploratory survey involving 1276 users across the United States, United Kingdom, and Australia established that online second-hand marketplaces have become widely utilised environment for buying and selling goods according to Quinn et al. (2023). The study further highlighted the increasing dependence on centralised digital systems to facilitate marketplace visibility and accessibility among users.

The growth of digital marketplaces has significantly influenced consumer behaviour and marketplace participation. Research on digital and online marketplaces shows that centralised systems improve the dissemination of information between sellers and buyers while reducing search-related inefficiencies as explained by Gu (2024). Such platforms enhance convenience by enabling users to access product information, seller availability, and marketplace activities from a single location. In addition, digital marketplace systems have been shown to improve transparency and accessibility within localised trading environments.

Within university environments, the informative market provides students with opportunities to purchase and sell items such as clothing, electronics, books and personal accessories at relatively affordable prices. These markets are particularly beneficial to students operating under financial constraints, as they create access to lower cost alternatives compared to conventional retail outlets. Informal student marketplaces also encourage use and redistribution of goods, contributing to more sustainable consumption practises within campus communities.

Despite the importance, university flea markets are commonly characterised by informal operational structures. Information concerning vendor schedules, product availability, and market location is often disseminated through word-of-mouth communications, social media posts, or temporary announcements. Research on digital mediated marketplaces suggests that the absence of centralised information systems limits accessibility and reduces marketplace efficiency within localised trading environments (Podkalicka & Fredriksson, 2023).

In academic institutions where students manage demanding schedules and multiple responsibilities, physically searching for the flea market activities may be inconvenient and time-consuming. Consequently, students may fail to access available opportunities for purchasing affordable goods and participating in campus space trading activities. Similarly, vendors may experience with this feasibility of their products due to inconsistent communication channels and limited reach among potential buyers.

The adaptation of digital marketplace platforms has demonstrated the ability to improve accessibility, organisation, and communication within trading environments. Research on platform-based marketplaces emphasises that digital system streamlines the interaction between sellers and buyers to centralising information and improving visibility of market activities as explained by Gu (2024). Furthermore, digital marketplaces support localised economic ecosystems by enabling users to access marketplace information conveniently through Internet enable technologies.

At Strathmore University, informatively market activities already exist and provide students with opportunities to exchange and acquire affordable goods within the campus community. However, the lack of a centralised and structured platform for free market information may reduce accessibility and participation among students. Therefore, there is a need to explore technological approaches that can improve the organisation, visibility, and accessibility of market activities within the university environment through a web-based marketplace system.

1.2 Problem Statement

Information about the Strathmore flea market is scattered and hard to find. Students usually hear about the sales through word-of-mouth or random social media posts. This creates a ‘gap’ where many people who might want to buy or sell items simply don’t know the market is happening. Ochieng and Mutua (2022) explain that when information is not maintained within centralised digital systems, student Communities experience difficulty supporting internal business activities.

Another major problem is that students are very busy. At Strathmore, students have to manage difficult classes, clubs and personal lives. Because of this, they have little free time to walk around campus just to see if a vendor is still open. Students within modern academic institutions increasingly require convenient and time efficient digital services due to demanding academic schedules and responsibilities. If a student does not know for sure that an item they need is available, they probably will not spend time looking for it. Instead, they might go buy something new and expensive, which is a missed opportunity to save money.

Finally, there is no way to check is for sale, how much it costs or where the vendor is located. This lack of clear information makes the market feel unorganized. Research shows that for any market to work well, people need to be able to trust that the information they see is accurate and easy to access as argued by Levin and Milgrom (2024). Without a website to list these details, student vendors lose on customers and buyers miss out on great deals. This keeps the campus from having a truly successful and sustainable way to trade goods among students.

1.3 Objectives

1.3.1 General Objectives

To develop a web-based flea market system to centralize information on student-led trade to improve accessibility, transparency and economic engagement within the university community of for Strathmore University.

1.3.2 Specific Objectives

- i. To identify the current informal methods students use to share and access flea market information within Strathmore University.
- ii. To investigate the specific challenges and market inefficiencies faced by students and vendors when utilizing these current informal systems.
- iii. To design the system architecture, user interface (UI) and database schema, and functional feature requirements for the proposed web-based marketplace platform.

- iv. To develop and implement the web-based flea market system using modern server-side and client-side web technologies.
- v. To test the developed web-based platform for usability, performance, and reliability within the university community.

1.3.3 Research Questions

- i. What current informal methods do students use to share and access flea market information within Strathmore University?
- ii. What are the specific challenges and market inefficiencies faced by students and vendors when utilizing these current informal systems?
- iii. How should the system architecture, user interface (UI), database schema, and functional feature requirements for the web-based platform be designed?
- iv. How will the web-based flea market system be developed and implemented using modern server-side and client-side web technologies?
- v. How will the developed web-based platform be tested for usability, performance, and reliability within the university community?

1.4 Justification

Flea markets within university environments provide students with access to affordable goods and create opportunities for small scale economic participation. However, the informant nature of these markets often limits accessibility and visibility of information related to available products, vendor schedules, and market locations. The proposed web-based flea market system seeks to address these challenges by centralising flea market information within a single digital platform accessible to students at Strathmore university.

Recent studies have shown that digital marketplace systems significantly improve information accessibility, transparency, and user convenience by reducing information as image between buyers and sellers as discussed by Veltri et al. (2020).

In addition, digital marketplace platforms have become increasingly important in improving the efficiency of commercial interactions by streamlining access to information and reducing such related inefficiencies (Bai et al., 2020). Bai et al. (2020) demonstrates that centralised systems improve seller visibility and simplify product discovery processes.

The proposed system will therefore benefit students by providing a structured and easily accessible platform through which they can identify available goods, vendor operating

schedules, and flea market locations without physically searching across campus. This will help students save time while improving awareness and participation in flea market activities.

Furthermore, the system will support student vendors by increasing their visibility of their products and activities within the university community. Steur and Seiter (2020) indicate that improving information flow and transparency contribute to greater marketplace efficiency and participation among users.

The proposed platform will also contribute to sustainability within the university environment by encouraging reuse and redistribution of goods among students. By improving access to second hand and affordable products, the system can promote more sustainable consumption practises while supporting the informal campus economy.

Finally, others said they will contribute academically by providing insight into the application of digital marketplace technologies within university based informal trading environments. The findings may serve as reference to for future studies related to digital marketplaces, student entrepreneurship, and informal economic systems in academic institutions.

1.5 Scope

The proposed system will be a web-based platform designed for use by students within Strathmore University. The system will allow vendors to upload and display information about items available for sale, including descriptions, categories, and availability. Students using the platform will be able to view listed items, search for specific products, and browse goods based on categories.

The system will provide information on vendor availability, including the days and times vendors will be present within the campus flea market. It will also display the location of flea market activities within the university to enable students easily identify where vendors are situated.

Additionally, the system will allow users to receive updates on newly available items and vendor activity. The platform will centralised flea marketing information, making it accessible to students at any given time through an Internet enabled device.

1.6 Limitations

The proposed system will not support online payment or transaction processes. All buying and selling activities will be conducted physically within the University.

The system will not include delivery or logic logistics services for purchased items. Users will be required to physically visit the Flea Market locations to complete transactions.

The platform will be limited to use within Strathmore University and will not support external users or vendors outside the university.

The system will require users to have access to an Internet enabled device in order to access the platform. Its performance will therefore depend on Internet availability and connectivity.

1.7 Delimitations

The proposed study will focus on the development of a web-based flea market system especially intended for use within the university. The system will target students participating in free market activities either as buyers or vendors within the university environment.

The proposed platform will concentrate on the management and dissemination of information related to flea market activities, including product listing, vendor availability, market schedules, and flea market locations. The study was therefore be limited to informational and organisational functionalities associated with compass flea market operations.

The system will only support web-based access and will not include the development of a dedicated mobile application. In addition, the platform will not incorporate advanced E commerce functionalities such as an online payment process, delivery services, or external commercial integrations.

The study will further be limited to flea market activities conducted within Strathmore University and will not extend to external marketplaces or commercial trading platforms outside the institution. Data collection and system evaluation will primarily involve students within the university community.

Finally, the project will focus on improving accessibility and organisation of flea market information rather than regulating pricing, vendor market policies, or financial transactions between users.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews scholarly literature associated with informal marketplace activities and digital marketplace technologies within educational and community-based environments. The chapter examines the present condition of flea market activities in university settings together with the approaches commonly used by students to obtain play market information. It also evaluates existing online marketplaces related to the proposed system, identifies gaps within the reviewed systems, and discusses the technologies relevant to the implication of the proposed platform. In addition, the chapter presents the conceptual framework guiding the study.

2.2 Informal Flea Market Operations in Strathmore University

Informal trading activities continue to play an important role within institutions of higher learning by supporting social and economic interactions among students. Within Strathmore university, flea market activities have increasingly become part of their student life, providing opportunities for students to buy and sell products such as clothing, electronics, books, and personal accessories at relatively affordable prices. This activity is great alternative avenues through which students can access goods conveniently within the campus environment while simultaneously supporting small scale entrepreneurial participation.

Ferraro et al. (2023) note that the growing popularity of second-hand economies and reus oriented marketplaces has largely been influenced by economic factors, sustainability concerns, and changing customer behaviour among young people and students (Ferraro et al., 2023). The adaptation of digital communication technologies has further transformed how informal marketplaces operate by improving the visibility and accessibility of products and vendors within localised trading environments.

Within Strathmore university, flea market activities are commonly organised informally in different locations around the campus depending on student participation and vendor availability. These flea market support students financially by providing affordable alternatives to former retail stores while creating opportunities for students to generate income through small scale trading activities. Maphosa and Maphosa (2022) suggest that students-centered informal economy is positively contribute to social interaction and economic participation within universities.

Despite their relevance within the university environment, flea market activities at Strathmore university remain largely decentralised and loosely coordinated. Information regarding available products, vendor schedules, and flea market locations is frequently shared through informal channels such as verbal communication, temporary posters, and social networking platforms. Although these approaches provide flexibility, they may also reduce consistency and accessibility of information among students.

Technological advancements in digital cameras have significant influence modern marketplace systems globally. Current marketplace environment increasingly relies on centralised digital platforms to facilitate communication and improve interaction between buyers and sellers. Gu (2024) explains that centralised marketplace systems improve accessibility of information and product visibility within digital trading environments. These developments have encouraged the adaptation of digital solutions aimed at improving organisation and efficiency within trading activities.

However, flea market activities within Strathmore university continue to rely on fragmented communication mechanisms that may limit awareness and participation among students. Due to demanding academic schedules and other responsibilities, students may have trouble consistently obtaining information about flea market activities or locating vendors and products within campus grounds. Consequently, interactions between buyers and sellers may not occur as efficiently as intended within university marketplace environments.

2.3 Existing Methods Used by Strathmore University Students to Access Flea Market Information

Students and vendors participating in free market activities within Strathmore University currently utilise different methods to obtain and share information related to available goods, vendor schedules, and flea market locations. These approaches are generally informal and vary depending on how vendors choose to communicate with potential buyers. Common methods include physical exploration of flea market locations, interpersonal communication among students, and the use of social media or messaging platforms like WhatsApp, Snapchat or even Instagram.

2.3.1 Physical Exploration

Physical exploration is one of the common approaches used by students within university to access information concerning flea market activities. Through this method, students move

around various campus locations in search of vendors, products, or ongoing flea market activities.

Studies examining and promoting systems indicate that face to face interaction continues to influence trust and participation within localised marketplaces (Ferraro et al., 2023). direct interaction between buyers and sellers therefore

However, reliance on physical exploration may reduce convenience for students operating under strict academic schedules period since flea market activities within the University may not always occupy fixed locations or predictable times, students may spend significant time attempting to locate products or vendors across campus. This may reduce efficiency and accessibility for participants involved in flea market activities within Strathmore University.

2.3.2 Word-of-Mouth Communication

Word-of-mouth communication is another commonly used approach through which students within Strathmore University obtain information about free market activities. Students frequently rely on friends, classmates, and social circles to receive updates regarding available products, vendor availability, and marketplace schedules.

Maphosa and Maphosa (2022) highlight the role of interpersonal communication networks in facilitating information exchange within localised economies. Information obtained through familiar social relationships is often perceived as trustworthy due to existing social interactions among students.

Despite these advantages, verbal communication presents several challenges within the university environment. Information may not reach all students equally commonly meeting awareness of available flea market opportunities across the campus community. Additionally, verbally shared information may sometimes be incomplete, inconsistent, or delayed therefore affecting accessibility and per perform participation in flea market activities.

2.3.3 Social Media and Messaging Platforms

The widespread use of digital communication technologies amongst university students has encouraged the ease of social media and messaging applications for sharing flea market information within Strathmore University. Apart from such as WhatsApp, Instagram, Facebook, and Telegram are commonly used by vendors and students advertise products, communicate vendor schedules, and announce free market activities.

Social media and messaging platforms provide relatively efficient communication channels that enable information to reach large groups of students quickly. Podkalicka and Fredriksson (2023) observe that social networking technologies improve product visibility and facilitate communication between buyers and sellers within digital marketplace environments.

However, these platforms are not specifically designed to support structured free market management within university environments. Information shared through social media groups or messaging platforms may become disorganised due to large volumes of messages, repetitive posts, and lack of centralised categorization. As a result, students University may encounter difficulties locating specific product listings or identifying current active vendors.

Although social media platforms improve communication compared to physical and verbal approaches, they may still fail to provide a dedicated and organised system specifically tailored for free market operations within Strathmore University.

2.4 Related Works

Many digital platforms currently exist that facilitate the buying and selling of goods through online systems. These platforms provide functionalities that support product advertisement, communication between users, and accessibility of marketplace information. This section examines selected digital marketplace systems relevant to the proposed web-based flea market intended for Strathmore University students.

2.4.1 Jiji Kenya

Jiji Kenya is a classified advertisement platform that enables users to advertise and browse listings for various products and services through categorised digital sections. The platform supports multiple categories including electronics, clothing, furniture, and personal accessories. Users communicate directly through the platform to negotiate transactions and arrange meetings.

The categorised structure of the platform improves accessibility and simplifies the process through which is users discover products. Veltri et al. (2020) explain that category is listing systems improve efficiency and accessibility of information among customers.

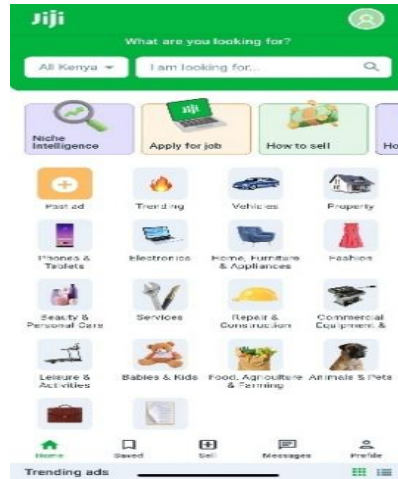


Figure 2.1: Jiji Kenya App Interface (Jiji,2026).

However, Jiji Kenya functions as a large-scale commercial platform serving diverse users across different geographical regions. As a result, the platform may not adequately support the localised communication and organisational requirements associated with flea market activities within Strathmore University.

2.4.2 Facebook Marketplace

Facebook Marketplace is an online marketplace feature integrated within the Facebook social networking platform that allows users to advertise and browse products within localised communities. Sellers can upload product images, descriptions, pricing information, and location details while buyers are able to research for products and communicate directly with vendors.



Figure 2.2: Facebook MarketPlace Interface (Facebook,2026).

The platform has gained widespread popularity due to its integration with within an already established social networking environment. Gu (2024) states that integrated online platforms improve accessibility and visibility of products within centralised digital environments.

Despite its effectiveness, Facebook Marketplace mainly serves broad public audiences rather than institution specific communities such as Strathmore University. Consequently, the platform may not sufficiently address the organisational and accessibility requirements associated with university flea market activities, including centralised campus vender coordination and student specific marketplace management.

2.4.3 eBay

eBay is a global ecommerce and online auction platform that facilitates the exchange of both new and second-hand goods through digital product listing. The platform enables users to upload product descriptions, images, and pricing information while supporting transactions through fixed price and auction-based mechanisms.

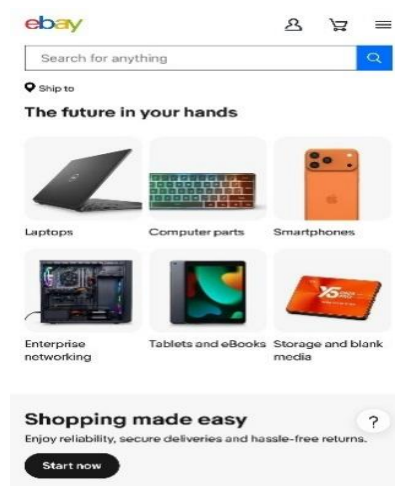


Figure 2.3: eBay Interface (eBay,2026).

The platform has significantly influenced the development of modern digital marketplace systems by demonstrating the effectiveness of centralised online environments in improving accessibility and organisation within commercial systems. Bai et al. (2020) explains that centralised ecommerce systems improve product visibility, information accessibility, and participation within digital marketplaces.

Despite these capabilities, eBay primarily supports formalised e-commerce operations involving online payments, shipping services, and large-scale commercial activities. These functionalities extend beyond the operational requirements of localised flea market activities

within Strathmore University, which mainly focus on improving accessibility and organisation of marketplace information within the campus environment.

2.5 Gaps in Related Works

Existing online marketplace platforms have introduced significant improvements in digital buying and selling activities by enhancing communication, accessibility of products, and interaction between users. Despite these advancements, several shortcomings still exist when such systems are evaluated within the context of localised university flea market operations such as the those found within Strathmore University.

Facebook Marketplace offers a digital environment that enables users to advertise products and communicate with potential buyers within nearby communities. Through its integration with social networking services, the platform improves product exposure and facilitates interactions among users (Gu, 2024). Nevertheless, the platform is primarily structured to support broad public trading activities rather than institution centred marketplace environments. Consequently, it lacks specialised functionalities required within university flea markets, including structured campus-based coordination of vendors, centralised scheduling of market activities, and organisation of flea market information specifically for students.

Jiji Kenya similarly provides an online classified advertisement environment where users can upload and browse categorise listings for various products and services. The platform simplifies product discovery by organising information into searchable sections and categories (Veltri et al., 2020). However, the system mainly serves a commercial audience distributed across multiple geographic regions. Because of this broad operational scope, the platform doesn't specifically address the communication and organisational needs associated with campus flea market activities within university communities.

eBay has played an important role in shaping modern E commerce systems by illustrating how centralised digital marketplaces improve accessibility of information, product visibility, and commercial participation among users (Bai et al., 2020). Despite these strengths, the platform mainly supports formalised commercial operations that involve online transactions, delivery systems, shipping logistics, and large-scale trading activities. Such functionalities extend beyond the requirements of flea market activities within Strathmore University, where marketplace interactions are largely localised and centred around physical exchange of goods.

Podkalicka and Fredriksson (2023) suggest that many digital marketplace platforms are designed for generalised commercial environments and may fail to support institutions-specific

communities. Consequently, these systems may not effectively address the specific organisation or challenges experienced within informal university-based flea markets.

Maphosa and Maphosa (2022) indicate that fragmented communication approaches negatively affect accessibility and efficiency within localised marketplaces. Within Strathmore University, flea market information is commonly distributed through verbal communication, physical exploration and social media interactions, approaches that may result in disorganised access to marketplace information among students.

Students within academic institutions frequently operate under demanding academic schedules and time limitations, reducing their ability to consistently monitor dispersed communication channels for flea market updates. Existing marketplace systems may therefore be insufficient in providing centralised and organised environment specifically tailored towards university flea market operations.

As a result, there remains a need for web-based flea market platform specifically designed for Strathmore University to improve organisation of marketplace activities, centralised flea market information, and enhanced accessibility of vendor and product information within the campus community.

2.6 Technologies to Be Used

This section outlines the technologies that will be used in the development of the web-based flea market system. This is to ensure that the system is robust, responsive and easy to maintain.

2.6.1 Infrastructure Deployment and Cloud Architecture (Render)

To transition the system from a development environment to a live, production ready platform accessible by the Strathmore University community, Render is adopted as the foundational cloud hosting infrastructure. Functioning as a specialized Platform-as-a-Service (PaaS), Render simplifies deployment by abstracting the underlying virtualization, networking configurations, and bare-metal server management. It establishes an automated continuously evaluating the project's centralized version-control repositories via secure webhooks, triggering instant software builds whenever code updates are committed. Within the core operational body of this multi-tier architecture, Render hosts the execution lifecycle of the backend scripting engine alongside its relational database clusters. Incoming user traffic from students on campus is handled through an integrated reverse-proxy layer that dynamically provisions Transport Layer Security (TLS/SSL) for end-to-end encryption. Furthermore, Render isolates internal system components within a private virtual network, shielding database transactions and

session states from external vulnerabilities. This robust cloud runtime environment guarantees high availability, rapid asset delivery, and a high responsive user experience for members of the Strathmore University community engaging with the digital marketplace.

2.7 Conceptual Framework

Students who are vendors register for a vendor account, upload or list their products and set their price. Students who are buyers log into the system using their Strathmore credentials, search items, view vendor information and check market location. The data is stored in a secure database where data can be stored or retrieved. The conceptual framework can be illustrated below;

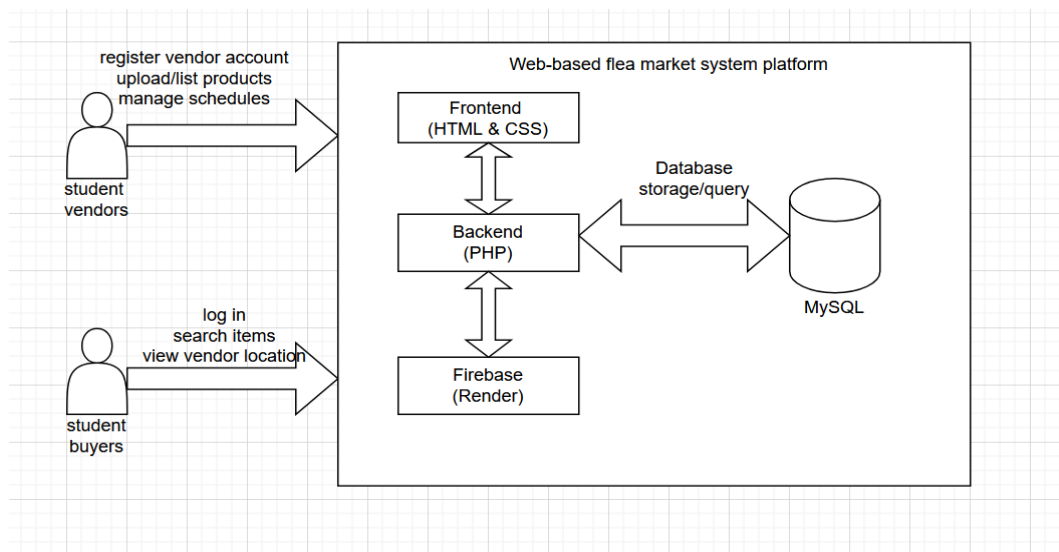


Figure 2.4: Conceptual framework.

Chapter 3: Methodology

3.1 Introduction

This chapter presents the methodology that will be used during the development of the proposed web-based flea market system for Strathmore University students. The chapter discusses the selected system development methodology, system analysis techniques, system design approaches, tools and technologies to be used during development, and the expected deliverables of the study.

3.2 System Methodology

system methodology refers to the structured process followed during the analysis, design, development, and implementation of a software system. The proposed web-based flea market system will adopt the Object-Oriented Analysis and Design (OOAD) methodology.

Pressman and Maxim (2019) describe OOAD as a software development methodology that focuses on identifying system objects, behaviours, interactions, and relationships during system development. The methodology is widely used in modern software engineering because it promotes modularity, reusability, scalability, and easier system maintenance.

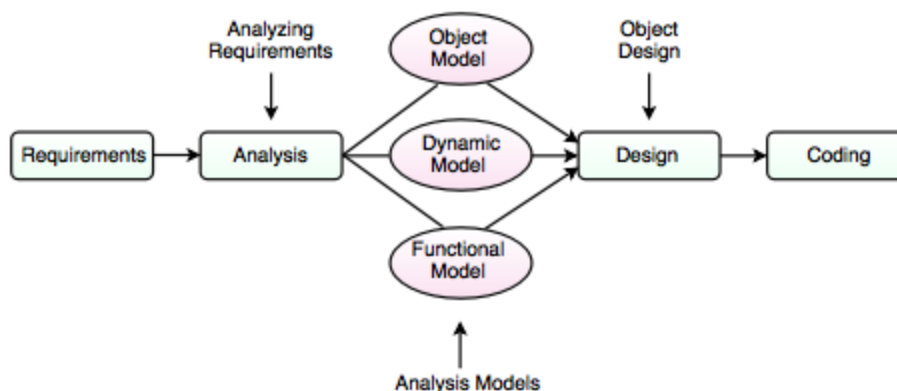


Figure 3.1 Object Oriented Methodology (CareeRide.com,2016)

The methodology will be appropriate for the proposed web-based flea market system because the system will involve different interacting entities such as students, vendors, administrators, products, categories, and flea market schedules.

The OOAD methodology will involve the following stages:

3.2.1 Requirement Analysis

This stage will involve gathering and analysing information concerning the current flea market environment within Strathmore University. Information will be collected regarding existing

methods used by students to access free market information and the challenges associated with those methods.

Functional or non-functional system requirements will also be identified during this stage. Functional requirements will describe what the system should do while non-functional requirements will describe how the system should operate.

3.2.2 System Analysis

During this stage come out the researchers will analyse the identified system requirements and determine how different users will interact with the proposed web-based flea market system.

System analysis stage will focus on identifying system users, user Interactions, data flow within the system, system processes, and relationships between different system entities

Analysis diagrams such as Use case diagrams, Sequence diagrams, and Entity Relationship diagrams will be identified and described during this stage.

3.2.3 System Design

The System design stage will involve creating the structure and architecture of the proposed system.

This stage will focus on database design, user interface design, navigation structure, system architecture, product management structures, and vendor management structures

The Design stage will help organise how Defences Team components will interact during implementation.

3.2.4 System Implementation

This stage will involve actual coding and development of the proposed web-based flea market system using selected tools and technologies.

The implementation process will involve developing functionalities such as product listing, product search, vendor information management, category organisation, flea market schedule management, and user interaction Interfaces

3.2.5 System Testing

System testing will be conducted to verify whether the developed system satisfies the intended requirements.

Testing activities will involve checking correct display of product information, proper functioning of search features, accuracy of stored information, navigation between webpages, and proper interaction between the database and user interface

System testing will help identify and correct errors before deployment of the proposed system.

3.2.6 Justification of the Selected Methodology

The Object-Oriented Analysis and Design methodology will be suitable for the proposed study because it supports modular development and simplifies management of complex system interactions.

The proposed web-based flea market system will contain multiple interacting entities including; students, vendors, products, administrators, and the flea market schedules. OOAD will therefore assist in organising these entities into manageable objects and classes.

Additionally, OOAD support system scalability and reusability, making it appropriate for future expansion of the proposed system (Sommerville, 2020).

3.3 System Analysis

System analysis involves examining the requirements and this is necessary for development of the proposed system. This section describes the analysis diagrams that will be used during analysis of the proposed web-based flea markets system.

3.3.1 Use Case Diagram

A use case diagram will be used to illustrate interactions between users and the proposed web-based flea market system.

The diagram will identify the primary users of the system such as students, vendors and system administrators.

The use case diagram will describe the activities performed by users including; viewing products, uploading product information, searching for products, managing flea market schedules, managing system listings.

According to Dennis et al. (2018), use case diagrams assist in identifying system functionalities from a user perspective.

3.3.2 Sequence Diagram

A sequence diagram will be used to illustrate interactions between system components overtime.

The sequence diagram will demonstrate how requests move between users, web pages, databases, and system processes during activities such as product uploads and product searches.

The diagram will assist in understanding communication flow between different components within the proposed system.

3.3.3 Entity Relationship Diagram

An entity relationship diagram (ERD) will be used to illustrate relationships between different data entities within the proposed web-based flea market system.

The ERD will include entities such as students, vendors, product, categories, and flea market schedules.

The diagram will assist in designing the database structure and identifying relationships between system entities.

3.4 System Design

This section provides details on the models that are meant to satisfy all the document requirements for the system. This section also demonstrates the input, output and processes meant to be performed by the system. These models are illustrated appropriately using wireframes, database schema and system architecture.

3.4.1 Wireframes

A wireframe is an outline of a user interface in a web page or application. It gives a visual representation of the layout, structure, functionality and user flow (Bruton,2022). It will be used to model the user interface of the proposed system, showing layouts for the landing page, product listings grid and vendor upload dashboard.

3.4.2 Database Schema

A database schema is a blueprint detailing both the physical and logical design of a relational database. It is defined by a set of rules that dictate the architecture of the database and the data collection needs. These rules provide structural constraints and relationship links for the user data, classified listings and vendor operational schedules within the proposed system.

3.4.3 System Architecture

System architecture is a fundamental concept in technology and engineering, as it details the structure and high-level design of a system (Raoque,2025). This will be used to break down the flea market

platform into a stable three-tier model consisting of a client browser presentation layer, a server-side application processing layer and a back-end data storage layer.

3.5 System Development Tools and Technologies

This section goes over the tools that are to be used in the development of the proposed system. Different development tools, frameworks and technologies are chosen based on their suitability for the proposed system.

3.5.1 HTML & CSS

HTML5 establishes the semantic structure and content layout of web applications, while CSS3 manages the presentation, typography and visual styling (Nixon,2021). These core web standards will be utilized together to create a clean, responsive user interface that adapt seamlessly to both desktop and mobile web browsers.

3.5.3 JavaScript

JavaScript is a lightweight, interpreted programming language used primarily to introduce client-side behaviour and dynamic interactivity to web interfaces (Flanagan,2020). Within the proposed platform, client-side scripts will handle tasks such as real-time form validation for item prices, interactive product filtering and interface state changes without forcing full-page reloads.

3.5.4 PHP

PHP is a mature, server-side scripting language designed specifically for web development to generate dynamic web content and process backend logic (Nixon,2021). PHP scripts will form the core application backend of the platform, securely processing HTTP requests, executing user session controls, handling item file uploads and driving communication with the database.

3.5.5 MySQL

MySQL is a widely adopted RDBMS that uses SQL to manage and query structured data tables (Ullman,2020). It will serve as the central data store for the flea market platform, maintaining strict relational integrity across tables for student credentials, active marketplace advertisements and vendor availability times.

3.6 System Deliverables

The successful completion of this project will result in a number of tangible deliverables spanning documentation and functional web system modules.

3.6.1 System Proposal

This documentation provides an overview of the web-based flea market platform to be developed. It explains why the system is needed, outlines its key functions and describes how it will solve current challenges. It also highlights the goals of the project and tools that will be used during development.

3.6.2 Authentication Module

Authentication ensures secure access to the platform. New users must register using their verified university credentials, after which they can log in. This module relies on server-side session tokens to differentiate between standard student buyers and student vendors while protecting account access.

3.6.3 Student Vendor Module

The student vendor module provides a backend dashboard layout where authorized student sellers can add, edit or delete classified advertisements. It allows vendors to upload item images, set price parameters, toggle listings statuses between available and sold and update their operational schedules on campus.

3.6.4 Student Buyer Module

The student buyer module is an interactive discovery interface engineered for optimal item search. It provides student consumers with functional tools to browse categorized item listings, apply keyword filters, view product parameters and check live schedules indicating when and where to meet a vendor physically.

3.6.5 Final Project Document

A primary academic deliverable of this study will be the Final Project Document. This comprehensive, formal report will synthesize the complete lifecycle of the project. It will encompass the refined introductory chapters, the literature review, and the detailed methodology from this proposal, while expanding to include chapters detailing the actual system implementation, code architecture, testing matrices, user evaluation results, and recommendations for future work. This documentation will serve as the final empirical record submitted for grading.

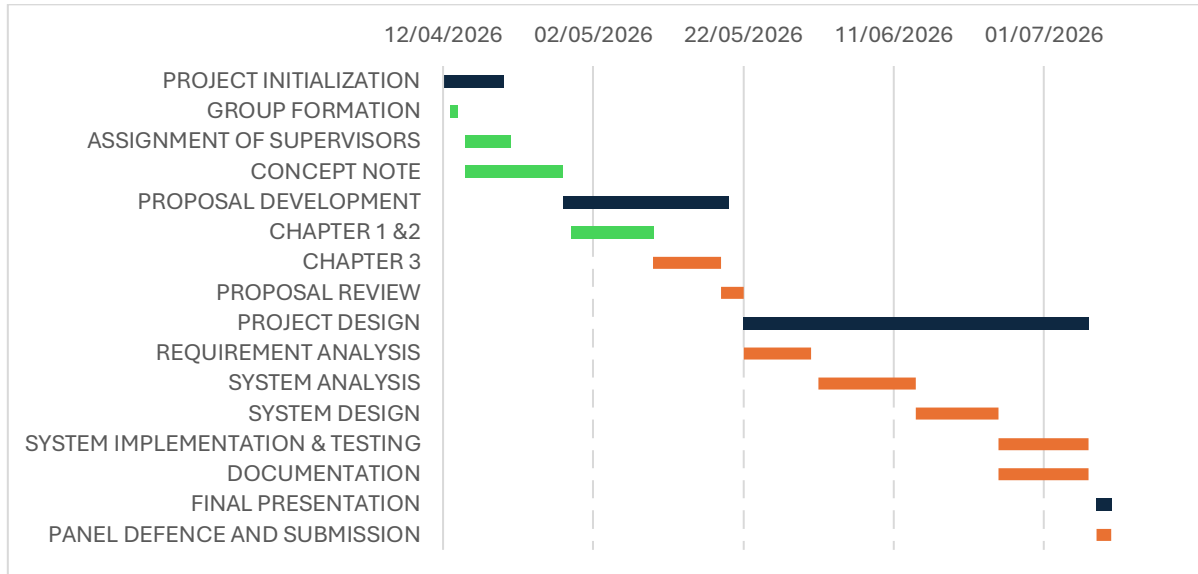
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Appendix

Appendix 1: Gantt Chart



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